



A Hybrid Deep Learning Framework for Stock Price Prediction Considering the Investor Sentiment of Online Forum Enhanced by Popularity

Huiyu Li¹ · Junhua Hu¹

Received: 3 June 2025 / Accepted: 23 December 2025

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026

Abstract

Accurate stock price prediction remains challenging due to the market's nonlinear dynamics and behavioral complexity. Existing models often neglect retail investors' sentiment and its amplification through popularity. This study proposes XSI-BiLSTM, a hybrid deep learning framework that improves stock price forecasting by incorporating retail investor sentiment. In this framework, we pioneer the use of XLNET to extract investor sentiment from stock market forums and, for the first time, integrate the resulting popularity-weighted sentiment indices with a BiLSTM-highway network for prediction, enabling the model to capture complex behavioral and temporal dependencies. Applied to the Chinese financial market, XSI-BiLSTM surpasses benchmarks by at least 8.15% (RMSE), 6.21% (MAPE), and 5.80% (R2). Validation and ablation experiments collectively confirm the contributions of popularity-adjusted sentiment and the highway mechanism to predictive performance of XSI-BiLSTM, jointly enhancing the reliability of forecasts. The findings empirically prove the predictive value of popularity-amplified sentiment and the potential of combining behavioral indicators with deep temporal learning to improve forecasting.

Keywords Stock price prediction · Investor sentiment · Post text analysis · XLNET model · BiLSTM model

✉ Junhua Hu
hujunhua@csu.edu.cn

Huiyu Li
huiyu_li@csu.edu.cn

¹ School of Business, Central South University, Changsha 410083, China

1 Introduction

The stock market functions as a dynamic marketplace for financial instruments and plays a pivotal role in the global economic ecosystem. Accordingly, accurate stock price forecasting is crucial for investors, analysts, and policymakers alike (Kanwal et al., 2022). However, stock price dynamics are governed by a complex, nonlinear interplay among determinants such as corporate performance, macroeconomic indicators, and investor psychology (Aljifri, 2023; Jin et al., 2020; Nikmah & Hung, 2024). This inherent complexity makes stock market forecasting one of the most challenging tasks in time-series analysis. Nevertheless, a substantial and growing body of empirical research shows that historical and multimodal patterns can be exploited to uncover meaningful predictive signals (Al-Khasawneh et al., 2025; An et al., 2024; Rezaei et al., 2021; Sharaf et al., 2021; Wang et al., 2024). These findings increasingly challenge the efficient market hypothesis, which posits that price fluctuations are purely stochastic (Fama, 1991).

Although significant progress has been made in forecasting stock dynamics, existing approaches remain limited. First, most studies still rely primarily on historical price series and technical indicators, while overlooking behavioral signals embedded in investor sentiment. Second, even when sentiment is considered, prior studies often simplify it into coarse binary categories—positive versus negative. Finally, sentiment-driven forecasting has largely focused on professional sources such as news, analyst reports, and announcements, leaving the collective sentiment of retail investors underexplored despite their growing influence (Sevinç & Coşkun, 2025).

These gaps are particularly salient given the rapid proliferation of online forums (e.g., Seeking Alpha, StockTwits, Eastmoney Guba), where retail investors actively share views that shape market sentiment. Drawing on behavioral finance and information cascade theory (Anderson & Holt, 1997; Bikhchandani et al., 1992), such collective expressions of sentiment may propagate across networks, reinforcing herding behavior and amplifying market fluctuations. Recent empirical evidence further supports this view, indicating that social media discussions embed predictive signals for subsequent stock movements (Costola et al., 2023; Huang et al., 2023,b; Pederesen, 2022). Unlike conventional numerical proxies, textual data from these platforms capture investor sentiment in real time, offering a more granular and direct reflection of market psychology (Eachempati et al., 2022). However, most studies overlook the heterogeneity of sentiment signals, which can significantly shape predictive performance.

To bridge these gaps, we focus explicitly on retail investor sentiment expressed in online forums rather than in professional news sources or analyst reports. We construct a refined sentiment index by fine-tuning XLNET to capture nuanced emotions in forum posts, further weighting these signals by engagement metrics. This design directly addresses two major limitations of prior research: the oversimplification of sentiment into binary labels and the neglect of popularity-driven amplification effects. The enriched sentiment signal is subsequently integrated with technical indicators and modeled using a highway-enhanced bidirectional long short-term memory (BiLSTM) network, enabling the framework to jointly capture behavioral sentiment

and temporal dependencies in stock dynamics. The contributions of this work are threefold:

- (1) We propose a hybrid framework that systematically integrates retail investor sentiment with technical and temporal features to enhance stock price prediction accuracy.
- (2) We empirically demonstrate that incorporating engagement metrics enhances the predictive utility of sentiment and captures the heterogeneity of sentiment signals that previous studies have overlooked.
- (3) Through comprehensive ablation studies on four representative stocks, we validate the effectiveness of the proposed framework and highlight the incremental value of both popularity-weighted sentiment indices and the highway-enhanced BiLSTM model in stock price forecasting.

In summary, this work systematically extracts retail investor sentiment from unstructured forum text and demonstrates its value for improving stock price prediction. The remainder of this paper is organized as follows: Section 2 reviews the relevant literature; Section 3 describes the proposed methodology; Section 4 outlines the experimental design; Section 5 presents and discusses the results; and Section 6 concludes with implications and directions for future research.

2 Related Work

In this section, we review prior research relevant to our study, focusing on three themes: text analysis on social media, time-series prediction in the stock market, and the relationship between investor sentiment and market performance. Finally, we provide a brief synthesis and discussion of the existing literature.

2.1 Text Analysis on Social Media

Human language is a representative form of unstructured data, and its analysis and understanding have progressed through four stages: rule-based, statistical, deep learning, and large-scale pretrained language models (LPLM) (Dong, 2023).

With the growing availability of LPLMs and increased computational power, large-scale analysis of online textual data has become feasible, establishing sentiment analysis as a key application in financial research. Prior studies have shown that news exerts significant influence on stock price returns (Li et al., 2014), and the analytical techniques used for this task have evolved from dictionary-based methods to LPLM-based approaches such as Transformers (Mishev et al., 2020).

Most existing studies, however, focus on financial news headlines because they are concise and information-dense (Johnman et al., 2018). With the enhanced ability of the LPLM to nonlinearly capture text features, some studies have begun to explore the use of multisource text data, including individual investor comments, for stock price prediction (Wu et al., 2022). More recently, scholars have examined investor-generated content, although its predictive role remains under debate (Daniel

& Titman, 1999). Furthermore, the interpretability of social media–derived text in the context of stock markets remains an active area of research (Gite et al., 2021).

Despite these methodological advances, existing studies share three key limitations. First, the overwhelming focus on professional news sources overlooks the behavioral heterogeneity of retail investors, whose sentiment may follow different dynamics than institutional actors. Second, binary sentiment classification (positive/negative) fails to capture the intensity gradations essential for understanding market impact. Third, prior work rarely accounts for the amplification effects of social engagement metrics, treating all posts as equally influential regardless of their reach or resonance within investor communities.

2.2 Time Series Prediction for the Stock Market

Stock market forecasting primarily relies on time-series techniques and fundamental analysis. These techniques can be broadly classified into three categories: regression-based, machine-learning-based, and deep-learning-based approaches.

Early studies on stock price forecasting predominantly employed traditional regression models, such as ARIMA, multiple regression (MR), and logistic regression (LR). For instance, Ariyo et al. (2014) applied ARIMA to stock data from two exchanges to build a prediction model, but the results still exhibited relatively large forecasting errors. Khan and Alghulaiakh (2020) demonstrated that a customized ARIMA model outperformed the automatic variant in predicting Netflix stock prices, although validation on a single stock limits its generalizability. Beyond ARIMA, other regression-based approaches, such as MR and LR, have also been applied to stock price prediction. Asghar et al. (2019) developed a multiple regression model that performed well across datasets through optimized predictor selection, although it required extensive preprocessing and dimensionality reduction. By optimizing the parameters of the LR model, Gong and Sun (2009) obtained better prediction results for the Shenzhen Development stock A than did the baseline model.

Despite their early contributions, regression-based approaches have inherent limitations. They rely on strong statistical assumptions such as stationarity, require extensive preprocessing and feature selection, and remain highly sensitive to noisy inputs. More critically, these linear models are ill-suited to handling the scale and heterogeneity of modern financial data, particularly unstructured modalities such as online investor text. Because real-world markets are volatile, nonlinear, and behaviorally driven, these constraints undermine both the robustness and generality of such models. These shortcomings underscore the inadequacy of traditional regression methods for contemporary forecasting tasks and motivate the shift toward machine learning and deep learning techniques, which can flexibly capture nonlinear relationships and leverage multimodal data at scale.

To overcome these limitations, researchers have employed machine-learning techniques to identify complex temporal features and improve the accuracy of stock price forecasting. Sapankevych and Sankar (2009) introduced the support vector machine (SVM) for stock market forecasting. Building on this, Shanthini et al. (2023) proposed a hybrid reptile search–remora–based SVM approach for more accurate stock movement prediction, underscoring the effectiveness of hybrid methods in stock market

forecasting. Park et al. (2022) proposed a multitask framework combining random forests (RF) for feature interpretation with LSTM models for predicting stock market returns. Such hybrid frameworks have become increasingly common in recent studies. Wang and Guo (2020) proposed combining the discrete wavelet transform with ARIMA and improving XGBoost to predict stock prices. Similarly, Yun et al. (2021) developed a GA-XGBoost framework for closing price prediction, though the marginal contribution of each component was not clearly articulated in their study. These hybrid frameworks demonstrate the importance of combining complementary strengths to handle high-dimensional numerical data and improve generalizability. However, despite significant advances in model depth and accuracy, most studies still adhere to the traditional time-series forecasting paradigm—focusing largely on technical indicators while overlooking external behavioral factors. This gap is precisely what current natural language processing (NLP) technologies can address and what our framework seeks to exploit.

2.3 Investor Sentiment and the Stock Market

Although earlier forecasting studies mainly focused on technical indicators, behavioral finance research has firmly established that investor sentiment exerts a significant influence on market dynamics. However, existing approaches to quantifying sentiment remain limited.

Early attempts to measure sentiment relied on aggregated numerical proxies such as volatility, turnover rate, and trading volume (Gong et al., 2022; Lin et al., 2024; Reis & Pinho, 2021). Although computationally convenient, these indirect proxies have two inherent shortcomings. First, they capture market outcomes rather than underlying investor psychology, thereby conflating sentiment with trading effects. Second, their aggregation obscures heterogeneity across investor segments—such as institutional versus retail, or informed versus noise traders—whose sentiments may influence markets in distinct ways.

Recent studies have begun extracting sentiment from textual sources, leveraging NLP advances. Valle-Cruz et al. (2022) demonstrated time-lagged correlations between Twitter sentiment polarity and global stock indices during the COVID–19 pandemic. Liu et al. (2023) used a convolutional neural network (CNN) to classify investor sentiment, verifying its synergy with stock prices and event sensitivity. Xu et al. (2022) found that managerial emotions expressed in corporate news releases predict returns during high-emotion periods; however, their focus on elite communications overlooks retail investor discussions. Other studies have also explored unconventional, non-textual signals. Obaid and Pukthuanthong (2022) extracted sentiment from news images to construct a market-level sentiment index that addresses the lack of visual information in financial markets. While innovative, this approach remains confined to professional media rather than extending to user-generated content. Luo et al. (2022) extended their analysis to livestreaming contexts and found that celebrity hosts generated stronger market reactions than key opinion leaders or corporate executives, suggesting that the impact of sentiment varies substantially by source and delivery mechanism. Other research exploring distant signals such as music con-

sumption patterns (Edmans et al., 2022) remains exploratory with limited theoretical grounding and unclear causal mechanisms.

Despite these advances, prior studies have largely focused on event-level sentiment, overlooking the fine-grained and continuous emotional expressions that characterize retail investor interactions on social forums, where day-to-day sentiment dynamics may exert a more consistent influence on stock movements.

Overall, three major research gaps remain. First, most text-based studies have concentrated on professional sources (e.g., news or analyst reports) rather than user-generated discussions. Second, sentiment effects are often assumed to be uniform, overlooking how social engagement and amplification mechanisms may shape their market impact. Third, existing frameworks typically model sentiment independently of technical market indicators, even though growing evidence suggests that behavioral and technical signals jointly enhance predictive performance (Gong et al., 2024).

2.4 Comparative Summary of Existing Approaches

As evidenced by prior literature, the integration of sentiment analysis into stock market forecasting has evolved from simple sentiment scoring to sophisticated deep-learning frameworks. Nevertheless, substantial divergence persists across studies with respect to data sources, model architectures, and sentiment representation strategies. To clarify the methodological heterogeneity and limitations of current research, Table 1 presents a comparative summary of representative studies conducted between 2021 and 2024, highlighting the methodological gaps that motivate our proposed framework. The comparison focuses on three key dimensions: modeling approach, sentiment data source, and feature integration strategy.

Methodologically, most prior research adopts deep-learning architectures for forecasting—predominantly LSTM models and their variants—owing to their ability to capture temporal dependencies in financial time series. In other words, the adoption of LSTM-based predictive modules in our study is well supported by prior literature.

The data sources employed in previous studies show a clear trend toward diversification. To ensure representative sentiment measurement, earlier works (e.g., Gite et al., 2021; Wu et al., 2022) primarily rely on financial news headlines, whereas more recent research (e.g., Das et al., 2024; Huang et al., 2023,b; Mu et al., 2023) has shifted toward user-generated content, such as Twitter posts and online investor forums. This shift underscores the growing recognition of the predictive value inherent in retail investor sentiment, which captures market psychology in real time. However, many existing studies still employ rudimentary sentiment analysis approaches—either simplifying emotional expressions into coarse polarity dimensions or failing to effectively align textual sentiment with technical indicators. These limitations constrain both the interpretability and predictive capability of such models. Notably, Das et al. (2024) is one of the few studies that explicitly consider sentiment heterogeneity. Nevertheless, our research differs from theirs in several key aspects. First, we incorporate sentiment extracted from both post headlines and body texts on investor forums, whereas Das et al. (2024) analyzed only the main body of Twitter posts. Our approach to sentiment aggregation also differs accordingly. Second, our framework employs the more advanced XLNET architecture, fine-tuned on

Table 1 Summary of representative studies on stock price prediction incorporating investor sentiment

Study	Method	Senti- ment data source	Feature	Limitations
Gite et al. (2021)	LSTM	financial news	stock prices of open, high, low and news headlines	Only news headlines were used, and there was no integration strategy with stock technical features; the failure to explain why the accuracy of LSTM-CNN was lower than that of the pure price model may indicate insufficient handling of textual sentiment.
Wu et al. (2022)	LSTM	stock posts and financial news	sentiment index, technical indicators and historical transaction data	Using only news headlines without the full text may result in the loss of deeper semantic meaning; relying on price fluctuations for sentiment labeling is somewhat tenuous, as it fails to account for the inherent emotional authenticity of the text.
Gong et al. (2022)	auto-regression distributed lag model	eight basic sentiment proxies	investor sentiment index	The sentiment index is aggregated and constructed from numerical proxies, without considering the actual expressions of individual users.
Lin et al. (2022)	LSTM with genetic algorithm	Twitter post	transaction and indicator data, sentiment scores	A multilingual stock market was considered, but Google Translate was directly used to translate into English; no targeted optimization of the model structure was performed for stock time-series forecasting; sentiment scores were directly divided into ten dimensions.
Huang et al. (2023, b)	LSTM, hybrid genetic algorithm	instant online discussion platform	five-dimensional sentiment indicators for comments and replies, stock technical indicators	The accuracy of binary classification is relatively low, and analyzing only a single stock results in insufficient generalizability.
Mu et al. (2023)	LSTM with sparrow search algorithm	investor forum	nine technical indicators and investor sentiment index	Using only the title text, sentiment classification is limited to “positive/negative,” with sentiment weights partially reliant on manual annotation, potentially affecting accuracy. The aggregation index calculation method is relatively simplistic.
An et al. (2024)	multi-modal ensemble model	news text and Baidu search volume	sentiment index and price	Simply concatenating low-dimensional attention scalars with high-dimensional semantic vectors overlooks the structural differences between them.
Das et al. (2024)	CNN, GRU and LSTM	Twitter post	five technical indicators, enhanced Twitter sentiment scores	Multiple text processing models were employed for ensemble sentiment analysis, yet the method of integration was not elucidated; distinguishing bot-generated comments in tweets was unfeasible; and the individual contributions of components within the framework remained unclear.
Liu et al. (2024)	LSTM-CNN, trellis network, attention mechanism	financial news	positive sentiment index and price	Only validate short-term forecasts; only test index predictions, without involving individual stocks

a domain-specific financial corpus to enhance sentiment extraction accuracy. Third, we conduct additional validation procedures to rigorously verify the robustness and explanatory power of our sentiment indices and model components. Despite these methodological differences, both studies reach the same conclusion: that engagement-enhanced sentiment indices can substantially improve the predictive accuracy of stock price forecasting models.

From a feature perspective, most recent studies, including ours, have moved beyond deriving investor sentiment from numerical proxies, instead combining textual sentiment features with traditional technical indicators. However, relatively few works have explicitly examined how to prioritize or balance the input of these two heterogeneous feature types within predictive models. Some studies, such as Liu et al. (2024), have attempted to address this issue through attention mechanism that assign dynamic weights to different inputs. Different from the mode of attention mechanism, our framework introduces a highway mechanism that enables the selective flow of information between sentiment and technical indicators. This design enables the model to adaptively regulate feature transmission, thereby achieving a more efficient fusion of diverse information.

3 Proposed Framework

3.1 Overview

Our study introduces XSI-BiLSTM, a hybrid framework for predicting stock prices. Figure 1 depicts the framework's two key components: the post sentiment analysis module and the prediction module.

As shown in Fig. 1, in the post sentiment analysis stage (upper part), online forum post data are preprocessed and analyzed by a fine-tuned XLNET to obtain sentiment scores for titles and body texts, which are then aggregated with popularity factors to construct a comprehensive sentiment index. In the stock price prediction stage (lower

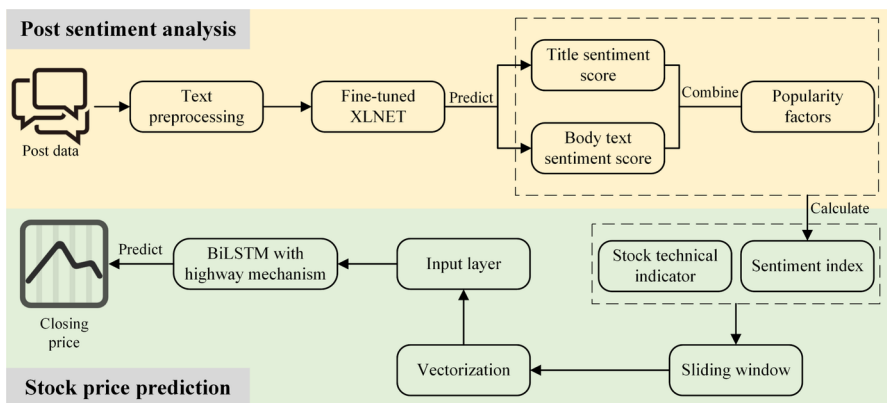


Fig. 1 Workflow of the proposed hybrid XSI-BiLSTM framework

part), the sentiment index and stock technical indicators are processed through a sliding window and vectorized, serving as inputs to a BiLSTM enhanced with highway mechanism to predict the next-day closing price.

To our best knowledge, this is the first study applying XLNET to financial forum posts for stock price prediction. Leveraging the Transformer-XL architecture (Dai et al., 2020), XLNET excels at processing extended text sequences, making it particularly effective for analyzing medium-to-long forum discussions. We fine-tuned XLNET on a relevant Chinese financial corpus to capture nuanced investor sentiment, which is essential for understanding the complexity of online discourse. Notably, our framework is the first to combine sentiment analysis with participation metrics for stock price prediction. This integration amplifies emotional signals and accounts for variations in sentiment intensity across posts. After obtaining sentiment representations from XLNET, these sentiment signals are combined with traditional stock technical indicators to form a comprehensive feature set. Then, these features modeled over time using the BiLSTM component, which captures temporal dependencies and sequential dynamics in stock prices. Our framework uniquely incorporates BiLSTM networks enhanced with highway mechanisms to address the complexity of stock price time-series prediction. The highway network mitigates the vanishing gradient problem, ensuring robust temporal information retention and improved accuracy, making our approach one of the first to apply this architecture to stock price forecasting.

3.2 Post Sentiment Analysis Based on XLNET

XLNET is a powerful autoregressive LPLM that achieves bidirectional context learning by randomly disrupting the order of sentences (Yang et al., 2019). It randomly adopts one of the orders, masks the end words, and then predicts the masked words autoregressively. Its objective can be formulated as follows:

$$\max_{\theta} E_{z \sim Z_T} \left[\sum_{t=1}^T \log p_{\theta}(x_{z_t} | x_{z_{<t}}) \right] \quad (1)$$

For a text sequence x of length T , there are $T!$ permutations, where z is one of all permutations Z_T . In addition, t represents the t -th element, and $<t$ represents the first $t-1$ elements. For example, consider the input text “[This stock has negative news]” and its random reordering “[has, stock, negative, This, news]”. The likelihood of optimization can be easily understood using Eq. (2):

$$p(z) = p(has) \times p(stock | has) \times p(negative | has, stock) \times p(This | has, stock, negative) \times p(news | has, stock, negative, This) \quad (2)$$

XLNET can learn to extract information about each alignment in a text sequence by sharing model parameters. However, it is impractical to compute all possible alignments due to computational constraints. Therefore, during pretraining, only

one alignment is randomly selected for each input sequence. Instead of destroying sequences, XLNET uses a mask matrix to implement the alignments. This approach maintains input format consistency between the fine-tuning and downstream tasks, thus avoiding discrepancies between pretraining and fine-tuning. To compensate for the corruption caused by the mask matrix, XLNET merges the position information into the softmax function likelihood as follows:

$$p_{\theta}(X_{z_t} = x | x_{z < t}) = \frac{\exp\left(e(x)^T g_{\theta}(x_{z < t}, z_t)\right)}{\sum_{x'} \exp\left(e(x')^T g_{\theta}(x_{z < t}, z_t)\right)} \quad (3)$$

where $g_{\theta}(\cdot)$ is the hidden state with z_t location information added. Additionally, XLNET employs a two-stream self-attention mechanism to train the model effectively. This mechanism involves two streams: the Query stream, which operates with only the current location information, and the Content stream, which has access to the current content information:

$$\begin{cases} \text{Query} : g_{z_t}^m \leftarrow \text{Attn}(Q = g_{z_t}^{m-1}, KV = h_{z < t}^{m-1}; \theta) \\ \text{Content} : h_{z_t}^m \leftarrow \text{Attn}(Q = h_{z_t}^{m-1}, KV = h_{z \leq t}^{m-1}; \theta) \end{cases} \quad (4)$$

In the above attention operation, Q , K , and V represent the query, key, and value, respectively. The content stream is set to the corresponding word embedding, i.e.,

$h_i^{(0)} = e(x_i)$. There are a total of $M = \{1, 2, \dots, m\}$ attention layers.

For objective (1) above, only the last token in the permutation sequence is predicted during training to avoid slow convergence. Specifically, z is split into a nontarget subsequence $z \leq c$ and a target subsequence $z > c$ by setting c as the cutoff point of the sequence. The objective is adjusted to:

$$\max_{\theta} E_{z \sim Z_T} [\log p_{\theta}(x_{z > c} | x_{z \leq c})] = E_{z \sim Z_T} \left[\sum_{t=c}^{|z|} \log p_{\theta}(x_{z_t} | x_{z \leq c}) \right]. \quad (5)$$

After training and fine-tuning XLNET, the sentiment tendency scores of the post were computed using the linear FC layer. The entire process of text prediction using XLNET is shown in Fig. 2.

In some natural language processing applications, such as sentiment analysis, XLNET shows promising performance, as demonstrated by our experiments. Therefore, we apply it to the text processing and analysis of posts. The specific process is outlined as follows: (1) Fine-tune XLNET using a relevant Chinese financial corpus to make it suitable for sentiment analysis of Chinese posts; (2) capture posts in web forums corresponding to the target stocks; and (3) use the fine-tuned XLNET to generate sentiment scores for the title and body of each post.

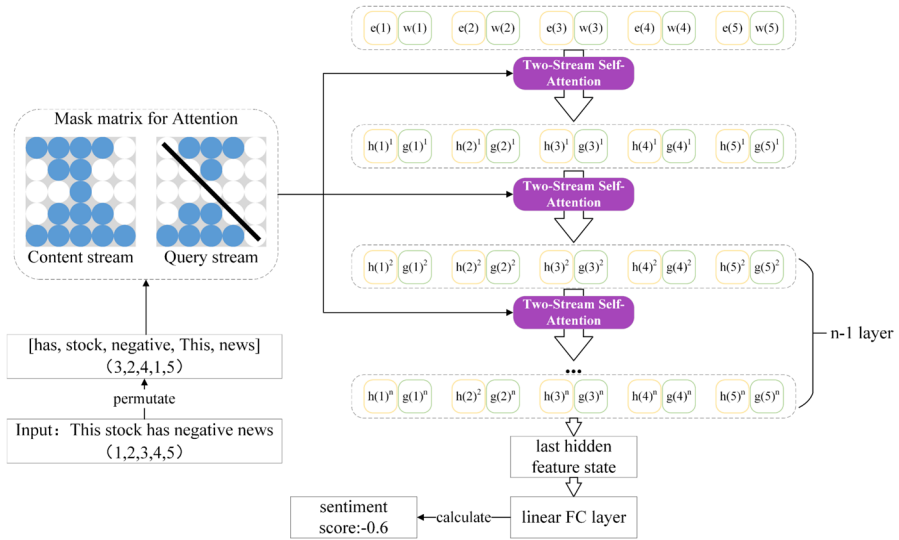


Fig. 2 Architecture of XLNET-based post sentiment analysis module. The input text is randomly permuted and processed through a mask matrix that separates the content and query streams (the content stream has access to its own tokens, while the query stream only receives positional cues). Each layer (hidden layer: h and query representation: g) applies two-stream self-attention to capture contextual dependencies. The final hidden feature state is passed through a fully connected (FC) layer to compute the sentiment score ranging from -1 (bearish) to $+1$ (bullish)

3.3 Construction of the Sentiment Index with the Popularity Factor

Considering the vast number of daily forum posts, feeding them directly into the predictive model would generate much noise. Therefore, it is necessary to aggregate the sentiment scores of the posts into a daily group sentiment index. Regarding the method of analyzing group sentiment, we first draw on the treatment proposed by Antweiler and Frank (2004) to compute the daily aggregated group sentiment index:

$$BI_t = \ln \left(\frac{1 + M_t^{bullish}}{1 + M_t^{bearish}} \right) \quad (6)$$

where BI_t represents the sentiment index for day t , $M_t^{bullish}$ is the bullish weight, and $M_t^{bearish}$ is the bearish weight. The original calculation method derives the weights by summing the related messages. However, this method is overly simplistic. Even when two posts share the same opinion about market fluctuations, they often differ in sentiment intensity. Therefore, it is necessary to calculate the weights in conjunction with the post sentiment tendency level. Given that the trained XLNET can provide the sentiment tendency score of the post, Eq.(6) is optimized as follows:

$$BI'_t = \ln \left(\frac{1 + \sum_1^m SBull_m^t}{1 - \sum_1^n SBear_m^t} \right) \quad (7)$$

where $SBull$ represents the propensity score of a post to be bullish for the stock market, with a value greater than 0, and $SBear$ represents the propensity score of a post to be bearish for the stock market, with a value less than 0. Eq.(7) replaces the previous daily group sentiment calculation method by summing the propensity scores of bearish or bullish posts.

Moreover, since user behaviors, such as reading, commenting, sharing, and liking, provide insights into users' perceptions and emotional inclination toward the post (Kim & Yang, 2017), we regard them as augmentations of the post's sentiment impact. Hence, we compute the daily group sentiment using Eq. (8), taking into account the amplification effect of the post's popularity factor on its sentiment:

$$BI_t' = \ln \left(\frac{1 + \sum_1^m SBull_m^t \cdot (R_m^{std} + C_m^{std} + L_m^{std})}{1 - \sum_1^n SBear_n^t \cdot (R_n^{std} + C_n^{std} + L_n^{std})} \right) \quad (8)$$

where R , C , and L correspond to the number of reads, comments, and likes on that post, respectively. Additionally, to reduce the impact of extreme values and accelerate the training speed, standardization is used on the popularity factor with Eq. (9):

$$x_{std} = \frac{x - \bar{x}}{\sigma} \quad (9)$$

where $\bar{x}$ is the mean of the original data and σ is the standard deviation.

To test whether the constructed sentiment indices more accurately reflect stock price trends, we apply the Granger causality test (GCT) (Granger, 1969) to examine their causal relationships. Since the post's title is exposed in the forum list and the body content is visible only after the visitors visit the post, we calculate their sentiment scores separately and name them the post title sentiment index (PTSI) and post body sentiment index (PBSI).

3.4 Stock Price Prediction Based on BiLSTM with a Highway Mechanism

LSTM is a type of classical RNN that effectively tackles the issues of gradient explosion and vanishing gradient within the training process. To further enhance the capability of time-series modeling of stock prices, we employ a bidirectional LSTM (BiLSTM) to integrate the contextual information. For related formula explanations, please refer to Appendix B.

Temporal data may suffer from gradient dispersion when entering a multilayer BiLSTM network, which ultimately hinders the network's learning efficiency. We introduced a highway mechanism to allow a seamless flow of information between the front and back layers (Zilly et al., 2017). To achieve this goal, we introduce a control mechanism of carry gates in the BiLSTM network to regulate the amount of information transferred from lower memory cell states to the next level. This is realized explicitly as follows:

$$d_t^l = \sigma (U_d x_t^l + W_d \otimes c_t^{l-1} + b_d^l) \quad (10)$$

$$c_t^l = d_t^l \odot c_t^{l-1} + f_t^l \odot c_{t-1}^l + i_t^l \otimes \tanh (U_c x_t^l + W_c h_{t-1}^l + b_c) \quad (11)$$

In the prediction module, we employ the improved BiLSTM-highway model described above, which effectively facilitates the flow of time series information from layer to layer. Figure 3 depicts the actual workflow of the module.

4 Experiment

4.1 Data Sources, Description and Processing

4.1.1 Textual Data

For our proposal, two types of textual data were required. The first type comprises Chinese investor stock comments with sentiment labels, which were used to fine-

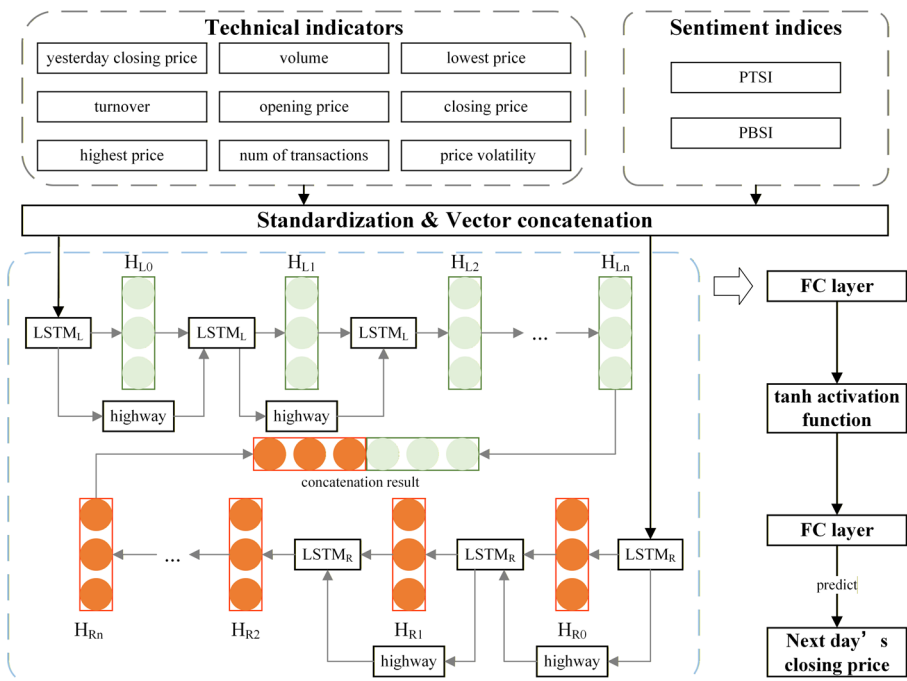


Fig. 3 The BiLSTM-highway module structure for stock closing price prediction. After standardizing the data, concatenate PTSI and PBSI with stock technical indicators into vector sequences. The combined sequences are processed by bidirectional LSTM layers, with one modeling the sequence from past to future and the other from future to past, while the highway mechanism ensures smooth information flow between layers. The last hidden states are passed through FC layers with a tanh activation function to predict the next day's closing price

tune XLNET. To construct this dataset, we combined four relevant corpora available on GitHub. Data sources: (https://github.com/algosenses/Stock_Market_Sentiment_Analysis,%20github.com/HongWooo/Sentiment,github.com/zyiyy/TextSentiment,github.com/zhy0313/double-degree). The sentiment labels in these corpora were created by using Chinese lexical segmentation and sentiment scores to categorize. Then, original researchers used model-based classification, which included LDA, RoBERTa, SVM, and ARIMA to verify accuracy of these labels. Notably, two of these datasets were collected from Eastmoney Guba, the largest financial online forum in China, which is also the source of the post data collected in our study. After summarizing the data and removing too short texts, 33,329 texts were obtained. A statistical description of the data is shown in Table 2. As shown, the proportions of positive (28.7%), neutral (38.9%), and negative (32.4%) categories are relatively balanced. The distribution ensures that the fine-tuned XLNET can effectively learn to distinguish.

The second dataset comprises forum posts extracted from Eastmoney Guba, capturing investor discussions with metadata including post titles, content, engagement metrics (likes, comments, and visits). We specifically selected discussions pertaining to two benchmark indices—the Growth Enterprise Index (SZ399006) and the SZSE Composite Index (SZ399001)—as well as two high-profile constituent stocks: Kweichow Moutai (SH600519) and Contemporary Amperex Technology (SZ300750). These securities were chosen based on their substantial investor following, which helps mitigate selection bias that could arise from focusing exclusively on less-traded equities. However, we note that the anonymized nature of forum

Table 2 Statistical description of the training data

Category	Text quantity	Example	Translation	Label
Positive	9554	没事解套很快的，耐心持有不用担心	It is okay. You will escape the encumbrance quickly; hold on patiently and do not worry.	1
Neutral	12,975	建议周一再观望一下	I suggest waiting and seeing again on Monday.	0
Negative	10,800	此股不建议，处于下降通道	Do not recommend this stock. It is in a downward channel.	-1
Total	33,329	—	—	—

participation precludes demographic stratification (e.g., by investor type, geographic distribution, or age). Consequently, our sentiment measures primarily reflect investor sentiment dynamics within online forums, extrapolation to broader market participants warrants careful consideration.

After collecting the initial post data, we performed a series of preprocessing steps using Python. First, the `replace_emoji` function from the `emoji` library was applied to remove color emojis, and regular expressions implemented in the `re` library were used to eliminate traditional emojis enclosed in brackets (e.g., “[smile]”). Second, hashtag-related content, including symbols such as “\$\$” and topics prefixed with “#” was removed using regular expressions. Third, we filtered out posts with empty titles or bodies, titles or bodies shorter than two characters, or bodies exceeding 500 characters, in order to reduce noise and improve representativeness. Fourth, to reduce duplication caused by web crawling, we sorted posts chronologically, then compared each title with that of the preceding post. If two consecutive titles were identical, the earlier entry was discarded. After preprocessing, approximately 340,000 posts collected between May 31, 2022, and May 31, 2023, were retained. A statistical description of the data is provided in Table 3, and an example post is shown in Fig. 4.

4.1.2 Technical Indicators

For the stocks introduced in Section 4.1.1, we selected nine technical indicators as their historical data for forecasting, as shown in Fig. 3. The above historical data

Table 3 Statistical description of the post data

Indicator	SZ399006	SZ399001	SH600519	SZ300750	Total
Post quantity	95392	55334	90144	96631	337501
Title average length	16.37	19.22	19.81	18.78	18.45
Body text average length	155.42	202.10	323.38	250.73	235.22
The average quantity of likes	5.17	7.03	2.73	3.24	4.27
The average quantity of comments	2.50	3.71	3.10	2.88	2.97
The average quantity of views	527.74	641.00	740.89	796.20	680.10



Fig. 4 Post example

were gathered from the RESSET database,¹ and the date range is consistent with that of the textual data, which are from May 30, 2022, to June 1, 2023. Due to the stock market being closed on weekends and holidays, approximately 247 days of actual trading history data were collected. Similarly, the technical indicators are standardized according to Eq. (9).

4.2 Metrics

4.2.1 Classification Evaluation Metrics of the Sentiment Analysis Model

We evaluate the performance of the models on the sentiment classification task using accuracy, precision, recall, and F1-score as evaluation metrics. The following equations define these metrics:

$$accuracy = \frac{TP + TN}{TP + FN + FP + TN} \quad (12)$$

$$precision = \frac{TP}{TP + FP} \quad (13)$$

$$recall = \frac{TP}{TP + FN} \quad (14)$$

$$F1 = \frac{2}{\frac{1}{precision} + \frac{1}{recall}} = \frac{2 \cdot precision \cdot recall}{precision + recall} \quad (15)$$

where *accuracy* is used to measure the ability of the model to classify samples correctly, *precision* represents the proportion of samples classified as positive cases that are positive cases, *recall* denotes how many of the samples predicted positively by the model were predicted correctly, and F1 is the harmonic mean of accuracy and recall, which may be a better metric when the categories are unbalanced.

In Eq.(12)–(15), “true positive” and “true negative” (TP and TN, respectively) relate to the ability to forecast the total number of positive or negative samples accu-

¹ Official website: <https://www.resset.com/>

rately. “False positive” and “false negative” (FP and FN) indicate the number of incorrectly predicted positive or negative samples, respectively.

4.2.2 Evaluation Metrics for Stock Price Forecasting

We employ standard regression model evaluation metrics, such as the root mean square error (RMSE), mean absolute percentage error (MAPE), and R2 score, to evaluate stock price forecasting tasks. Their expressions are shown as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (16)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{(y_i - \hat{y}_i)}{y_i} \right| \quad (17)$$

$$\text{R2} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (\bar{y}_i - \hat{y}_i)^2} \quad (18)$$

In Eq.(16)–(18), n is the total number of samples, y_i and $\hat{y}_i$, respectively indicate the actual and forecasted stock price, and $\bar{y}_i$ indicates the stock’s average price in the testing set.

4.3 Experiment Settings

4.3.1 Benchmark

To rigorously evaluate the proposed framework, we compared it with a set of widely used and conceptually diverse benchmarks. Support Vector Regression (SVR) was chosen as a strong ML baseline for time-series regression (Beniwal et al., 2023). RNNs were chosen as a baseline architecture owing to their efficacy in capturing sequential dependencies. Within this framework, we incorporated BiLSTM networks which leverage contextual information, alongside two attention-enhanced variants—LSTM-Att and GRU-Att (Althelaya et al., 2018; Niu & Xu, 2020; Zhang et al., 2019). These models integrate attention mechanisms to dynamically prioritize segments of the input sequence, thereby optimizing performance in forecasting tasks. Finally, the Transformer model (Vaswani et al., 2017) was selected. Unlike normal attention mechanisms, Transformer’s self-attention mechanism processes the entire sequence simultaneously. While it has shown promising results in various natural language tasks, its application to time-series forecasting still holds unexplored potential. Together, these baselines allow for a comprehensive assessment of the proposed framework against both classical and modern deep learning approaches.

4.3.2 Parameter Selection

The hyperparameter settings of XSI-BiLSTM and other benchmarks are provided in Table 4. These settings were selected based on prior research, computational constraints, and performance optimization.

To balance training efficiency, a uniform batch size of 32 was set across all prediction models. Hyperparameters, including learning rate (5×10^{-5} , 1×10^{-4} , 1×10^{-3} , 1×10^{-2} ,

Table 4 Hyperparameter settings of XSI-BiLSTM and other benchmarks

Module/Model	Hyperparameter	Parameter
Post-sentiment analysis module of XSI-BiLSTM (Based on XLNET)	Batch size	16
	Max length	128
	Epochs	100 with earlystop
	Warm up ratio	0.1
	Learning rate	2×10^{-5}
	Dropout	0.2
	Batch size	32
	Hidden layer size	128
	Learning rate	1×10^{-3}
	Weight decay	1×10^{-2}
Prediction module of XSI-BiLSTM (Based on BiLSTM)	Epochs	1000 with earlystop
	Batch size	32
	Hidden layer size	128
	Learning rate	1×10^{-3}
	Weight decay	1×10^{-2}
LSTM-Att/GRU-Att	Epochs	1000 with earlystop
	Batch size	32
	Hidden layer size	128
	Learning rate	1×10^{-3}
	Weight decay	1×10^{-2}
SVR	Epochs	1000 with earlystop
	Gru_layers/Lstm_layers	1
	Num_att_heads	8
	C	2
	Cache_size	200
	Degree	3
	Epsilon	0.1
	Kernel	linear
	Tol	1×10^{-3}
	Batch size	32
Transformer	Init_range	0.1
	Dropout	0.05
	Num_encoder	2
	Learning rate	1×10^{-3}
	Weight decay	1×10^{-2}
	Epochs	1000 with earlystop
	Positional encoding	sinusoidal

1e-1]), weight decay ([1e-3, 5e-2, 1e-2, 1e-1]), number of hidden size ([8, 16, 32, 64, 128, 256]), and number of attention heads ([2, 4, 8, 16]), were optimized using grid search within these predefined ranges, informed by prior research and established best practices. For models like LSTM-Att and GRU-Att, the optimal BiLSTM configuration was used as a baseline. Non-tunable and second-important parameters, such as the positional encoding type for Transformer and dropout rate, were set according to original paper settings. For SVR, we experimentally found that linear kernels outperform other kernels, so we keep the linear kernels and use the default scikit-learn settings for the other parameters. To prevent overfitting, all training processes incorporated early stopping mechanisms.

4.3.3 Training Process

To ensure the validity and reliability of our experiments, we separated the training and testing datasets for both the text and stock data. For the text data, we split it into training, testing, and validation sets at an 8:1:1 ratio. Cross-entropy was used as the loss function during training, as shown in Eq. (19).

$$L = - \sum_{i=1}^m p(x_i) \log(q(x_i)) \quad (19)$$

where $p(x_i)$ denotes the actual label with a value of 1 or 0. $q(x_i)$ is the probability of belonging to the category as output by the model after softmax computation, and m is the number of samples.

For stock price prediction, we adopted an 8:2 training-to-testing ratio, leveraging historical data from the preceding 7, 15, and 30 days as input windows for the model. The mean squared error was utilized as the loss function to quantify training accuracy.

5 Results

5.1 Training and Comparison of Sentiment Analysis Models

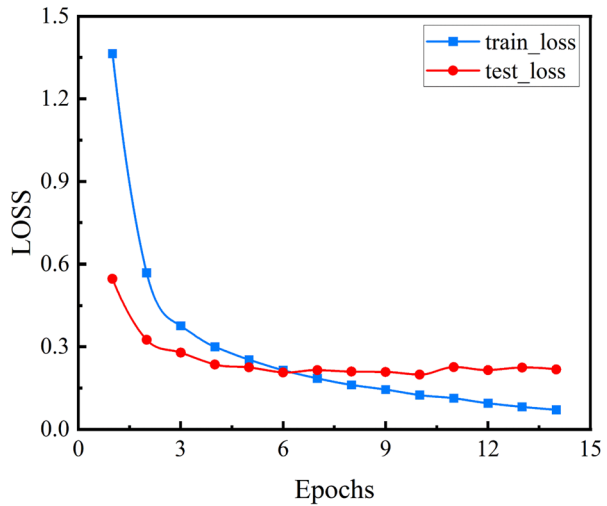
Table 5 presents the performance results of five sentiment analysis models, including XLNET, SVM, Naive Bayes (NB), BERT, and TextCNN. Among the models tested, XLNET achieves the highest F1-Score of 81.97, indicating its superior performance in sentiment analysis. BERT follows closely with an F1-Score of 81.68, showing that language pre-training models (LPLMs) like XLNET and BERT outperform traditional machine learning models such as SVM and NB. This performance trend confirms the advantages of LPLMs over traditional approaches, aligning with findings from previous research (Han et al., 2021). Additionally, XLNET demonstrates a rapid convergence rate in transfer learning, as illustrated in Fig. 5, where the training and testing loss stabilizes after just a few epochs.

The comparison results show that applying XLNET to analyze the hidden sentiment in textual data is effective, and its performance is the same as that of BERT.

Table 5 Performance results of five sentiment analysis models. The best results are highlighted in bold, followed by the same

Metrics	XLNET	SVM	NB	Bert	TextCNN
Accuracy	81.64	44.97	74.20	81.34	76.12
Precision	82.73	45.07	77.20	82.33	77.06
Recall	81.49	44.98	72.79	81.24	75.47
F1-Score	81.97	44.52	73.43	81.68	75.99

Fig. 5 Changes in loss during the training process of XLNET over time



Considering that the structure of XLNET is more suitable for processing long text (Sweidan et al., 2021), it is reasonable to believe that XLNET can better analyze the sentiment of a post, which often contains hundreds of words in the body of the post, compared to other models.

Then, we use the fine-tuned XLNET to analyze the post sentiment and output scores through the final FC linear layer, with a range of $[-1, 1]$. When the score is closer to -1 , 0 , or 1 , the post's sentiment toward stock prices tends to be bearish, neutral, or bullish, respectively.

5.2 Sentiment Validity Verification

In this section, we present the results of the GCT firstly to showcase the effectiveness of utilizing the sentiment index in predicting stock closing prices. We also analyze whether sentiment indices better capture stock market fluctuations when incorporating the popularity factor. Appendix A presents the results of the augmented Dickey-Fuller (ADF) test for both the price rate of change (ROC) and the sentiment index, both of which demonstrate significance. The GCTs with different lag durations are shown in Table 6. Among these indices, the PTSI and PBSI are sentiment indices that do not consider the popularity factor, whereas the PTSI' and PBSI' are considered.

As illustrated in Table 6, fluctuations in stock prices emerge as a pivotal determinant influencing variations in user sentiment within online forums, with all indices demonstrating significance at the 1% level across differing lag orders. This suggests

Table 6 Results of GCTs with lags of 1, 2, and 3 days

Stock code	Variables	Sentiment index does not Granger cause ROC			ROC does not Granger cause senti- ment index		
		lags(1)	lags(2)	lags(3)	lags(1)	lags(2)	lags(3)
SZ399006	PTSI	0.79	1.13	0.48	127.64***	54.79***	25.51***
	PBSI	0.48	2.56*	0.94	180.17***	79.59***	31.91***
	PTSI'	6.01**	3.38***	0.62	112.25***	48.24***	22.61***
	PBSI'	0.82	2.58*	0.75	172.73***	80.69***	34.56***
SZ399001	PTSI	0.33	0.33	0.04	132.69***	56.97***	23.66***
	PBSI	0.19	0.06	0.08	172.41***	67.17***	25.52***
	PTSI'	2.77*	2.85*	1.22	96.32***	79.81***	46.77***
	PBSI'	1.56	0.43	0.54	122.75***	87.27***	48.87***
SZ600519	PTSI	0.09	1.85	0.17	103.41***	39.20***	15.89***
	PBSI	0.65	1.98	0.92	134.45***	57.27***	24.06***
	PTSI'	3.67*	1.16	1.62	81.83***	62.57***	30.93***
	PBSI'	0.08	0.60	0.81	99.15***	91.85***	41.83***
SZ300750	PTSI	1.38	0.17	0.82	222.91***	108.06***	47.50***
	PBSI	1.42	0.48	1.67	324.65***	157.37***	79.07***
	PTSI'	2.14	1.61	2.13*	87.73***	87.25***	24.52***
	PBSI'	3.04*	0.69	0.60	227.66***	228.34***	56.50***

***, **, and * represent significance at the 1%, 5%, and 10% levels respectively, followed by the same

that the sentiment embedded within posts serves as an effective barometer of users' reactions to stock price movements.

In contrast, PTSI and PBSI demonstrate weaker Granger causality effects on stock price fluctuations. Only in the case of SZ399006 at a lag of 2 does PBSI attain significance at the 10% level. However, upon integrating the popularity factor, both PTSI' and PBSI' achieve significance at specific lag intervals across the four stocks examined.

This can be attributed to the heterogeneity of information shared within online forums, where diverse and sometimes conflicting opinions may obscure collective judgment unless filtered effectively. By integrating the popularity factor, the influence of discussions that resonate most strongly with investors is amplified. This adjustment enhances the sentiment index's ability to represent the aggregate and authentic market mood, rather than random or fringe opinions. From a behavioral finance perspective, this mechanism aligns with theories of social amplification of sentiment (Doroshenko & Tu, 2023) and attention-driven trading (Barber et al., 2022). Popular posts act as behavioral "attention cues" that guide market participants' collective interpretation of financial news, triggering herding and information cascade effects. Consequently, sentiment signals amplified through popularity metrics do not merely reflect reactions to past prices—they actively contribute to the formation of future expectations and price adjustments. The stronger impact of PTSI' relative to PBSI' further supports this interpretation. Because post titles are prominently displayed and often determine whether users engage with a discussion, they serve as an initial filter of investor attention. This corresponds to the availability heuristic in behavioral economics, where easily accessible or salient information disproportionately influences decision-making. Therefore, the PTSI' index, which captures title-based and popu-

larity-weighted sentiment, better represents the dominant expectations and emotional tendencies of retail investors than the PBSI’.

To further illustrate the impact of sentiment indicators on stock market returns, Fig. 6 depicts the average returns under different sentiment conditions. Sentiment indices with normalized absolute values greater than 0.5 are categorized as strong sentiment, while the other is classified as weak. The results show that the magnitude of average returns corresponds closely to the level of sentiment certainty reflected in the posts. After incorporating the amplification effect of popularity, the differences in return compared with the PTSI and PBSI are not substantial; however, instances where stock returns contradict the expected sentiment direction become less frequent, indicating a more robust alignment between collective sentiment and subsequent market outcomes.

When a dominant narrative gains influence through repeated dissemination in online forums, investors are likely to adjust their expectations accordingly, leading to short-term deviations from fundamental values, which is consistent with herding theories. Investors should recognize that when conflicting emotional discussions arise, stock market expectations become more unpredictable (as demonstrated by the weakly pessimistic scenario presented by PBSI). At such times, they should pay closer attention to whether the widely disseminated sentiments in posts receive consistent attention and recognition, rather than merely focusing on the volume.

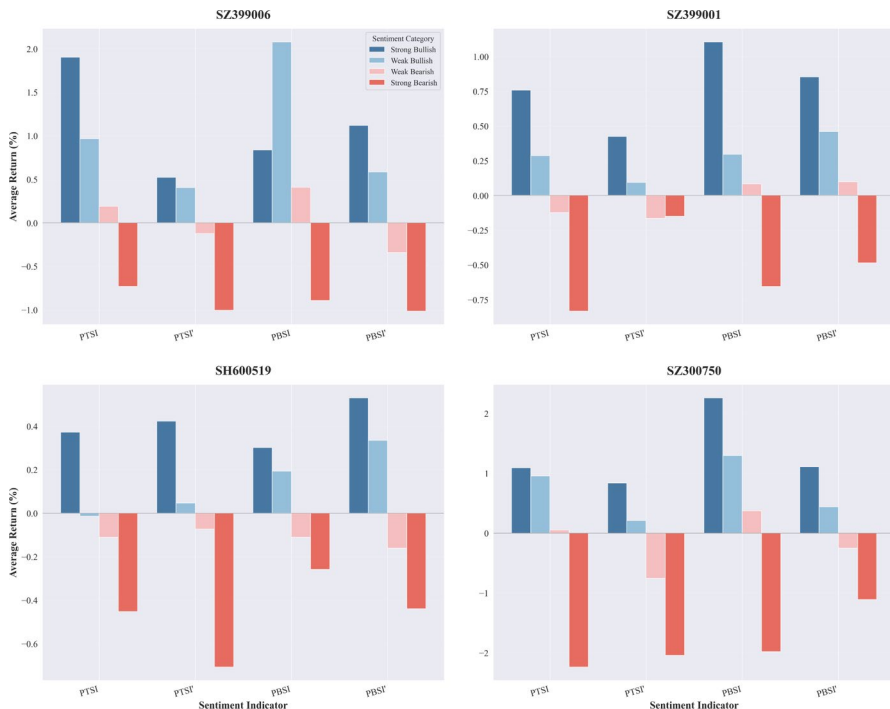


Fig. 6 Average stock returns under different investor sentiment categories

In summary, PTSI' and PBSI' capture forward-looking investor sentiment toward closing prices, thereby serving as reliable indicators for stock price forecasting.

5.3 Prediction Results and Comparison

The predicted prices of the models are compared with the actual stock prices using the introduced metrics to evaluate the effectiveness of the XSI-BiLSTM framework. Table 7 presents the mean performance metrics of each model, averaged across various input window sizes, together with the results of paired t-tests.

Notably, the XSI-BiLSTM model demonstrates superior efficacy across all metrics. Specifically, XSI-BiLSTM outperforms the second-best models by at least 8.15% (SZ300750), 6.21% (SZ300750), and 5.80% (SH600519) in terms of overall performance on RMSE, MAPE, and R2, respectively. Meanwhile, all model performances exhibit significant differences compared to XSI-BiLSTM. This preliminary finding underscores the robustness and efficacy of the proposed framework. To observe the improvements of the model under different scenarios more clearly, Fig. 7 is plotted.

As shown in Fig. 7, XSI-BiLSTM achieves great improvements over the mean performance of models in all cases. An intriguing observation is that increasing the input window duration does not yield a corresponding enhancement in predictive accuracy across the models. In fact, the SVR model has seen a decline in performance, probably because stock prices are instability, and simple linear kernel functions are not complex enough to handle long non-stationary time series data information. In contrast, XSI-BiLSTM consistently demonstrates outstanding predictive performance across both short-term and long-term input windows. This phenomenon can be attributed to its ability to bidirectionally model long time series and dynamically integrate real-time fluctuations in investor sentiment. However, it is noteworthy that XSI-BiLSTM did not demonstrate significant performance advantages across all metrics for SZ300750. To investigate the reasons behind this, we plotted the closing

Table 7 Comprehensive performance comparison of different models

Stock	Metrics	SVR	GRU-Att	BiLSTM	LSTM-Att	Transformer	XSI-BiLSTM
SZ399006	RMSE	48.56	51.99	47.58	52.07	68.40	37.34
	MAPE	1.69	1.81	1.68	1.83	2.56	1.34
	R2	0.58	0.53	0.61	0.53	0.19	0.75
SZ399001	RMSE	237.46	171.52	120.45	170.85	169.39	100.12
	MAPE	1.73	1.28	0.81	1.22	1.22	0.69
	R2	0.55	0.77	0.89	0.76	0.77	0.92
SH600519	RMSE	27.54	28.13	23.48	29.85	33.01	20.64
	MAPE	1.25	1.33	1.07	1.39	1.55	0.91
	R2	0.58	0.56	0.69	0.50	0.39	0.73
SZ300750	RMSE	6.41	4.88	4.54	4.56	5.45	4.17
	MAPE	2.32	1.76	1.65	1.61	1.94	1.51
	R2	0.19	0.53	0.59	0.59	0.40	0.68
Mean difference		1.8892	1.2009	0.5197	1.1746	2.0040	—
p value		6.1376	7.4122	6.9647	4.7371	7.0366	—
significance		***	***	***	***	***	—

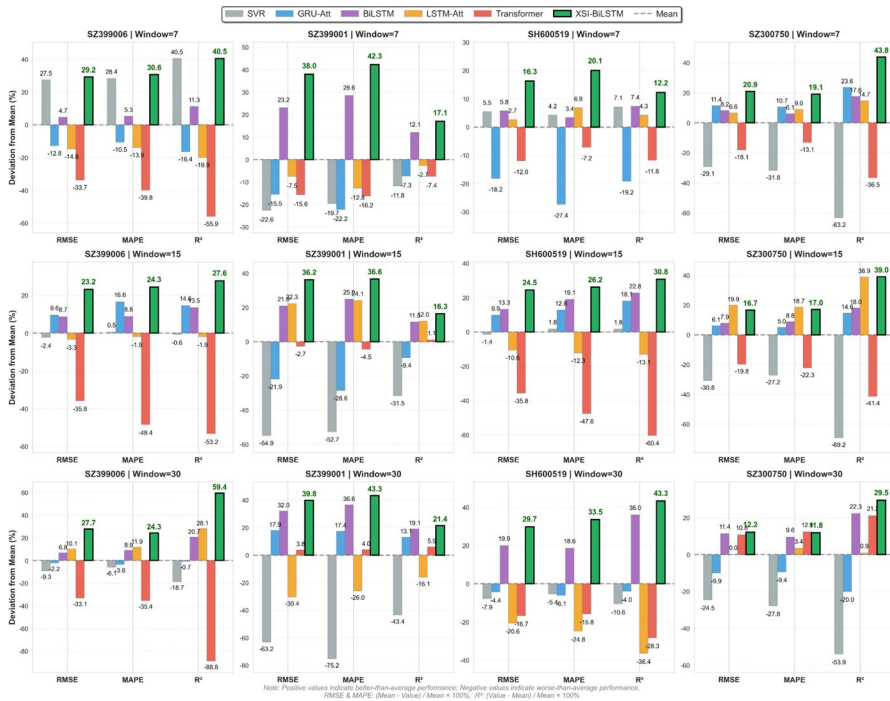


Fig. 7 Performance improvement of models across different time windows

price predictions on the testing set (March 21, 2023 to May 31, 2023) of each model against the actual closing prices in Fig. 8.

As illustrated in Fig. 8d, SZ300750 exhibits frequent range-bound fluctuations lacking discernible directional trends, contrasting with other stocks demonstrating clear downward trajectories. This heightened volatility disrupts the temporal dependencies crucial for bidirectional modeling of stock prices and sentiment, thereby undermining model stability. This phenomenon is further evidenced by the diminished performance of BiLSTM, which loses its status as a suboptimal predictor.

Several factors contribute to this limitation. First, before adequately capturing underlying periodic structures, complexity of XSI-BiLSTM coupled with early-stop mechanisms may lead to premature convergence on short-term patterns. Second, during market fluctuations, investor sentiment often shows obvious short-term fluctuations, which are exacerbated by arbitrageurs' mispricing (Lee et al., 2002), making it more difficult to predict stock prices based on emotions. In contrast, attention-based architectures (e.g., LSTM-Att and Transformer) demonstrate superior adaptability by dynamically weighting recurrent temporal features, thereby identifying stable patterns. Nevertheless, XSI-BiLSTM remains competitive within the top performance tier. Future refinements incorporating adaptive emphasis on cyclical timing and sentiment dynamics could further enhance its predictive robustness.

When examining the prediction results in more detail, as shown in Fig. 8, XSI-BiLSTM consistently exhibits stable tracking performance across the four stocks, with minimal bias. In contrast, SVR shows initial accuracy but gradually diverges

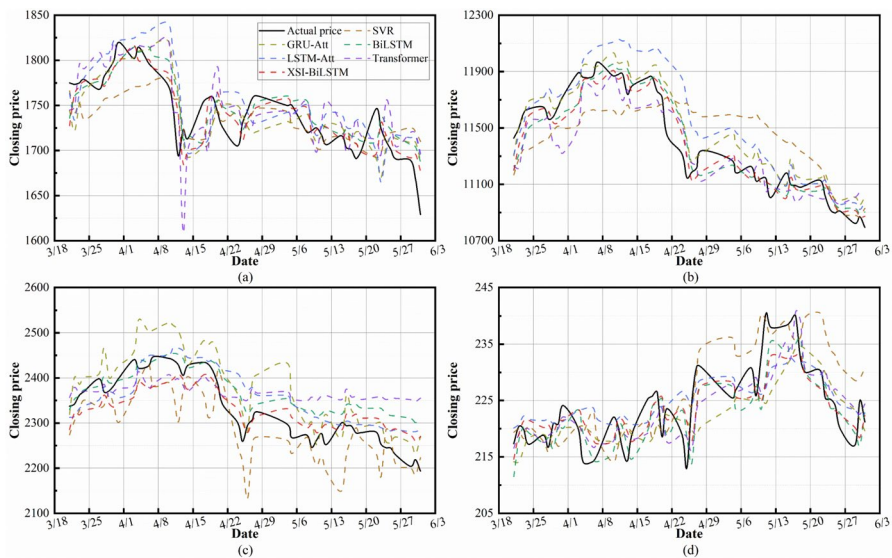


Fig. 8 Actual closing price and predicted price of four stocks under different models. **a:** SZ399006 **b:** SZ399001 **c:** SH600519 **d:** SZ300750

due to its limited capacity to model non-stationary dynamics beyond the training horizon. GRU-Att and LSTM-Att achieve good results for certain stocks yet display instability in others, possibly resulting from suboptimal attention weight distribution. The Transformer tends to overemphasize short-term fluctuations, leading to exaggerated responses to abrupt market changes. BiLSTM demonstrates superior trend-following capability compared to these models; however, its absence of sentiment feedback hinders rapid adaptation to market shifts. Collectively, these results highlight that integrating sentiment information allows XSI-BiLSTM to achieve a more balanced response between stability and sensitivity in stock price forecasting.

Figures 9, 10, 11, and 12 show each model's relative percentage error (RPE) and average one-sided error (AOSE) between the predicted and actual closing prices. Comparatively, BiLSTM, Transformer, and XSI-BiLSTM exhibit relatively small and stable overall biases. However, the RPE of the Transformer is more severe on certain trading days, e.g. April 13, 2023 and April 20, 2023 for SZ399006; May 11, 2023 for SZ300750. In some cases, the GRU-Att and LSTM-Att models are too "optimistic" and predict stock prices much higher than their actual values. This likely arises from the models' tendency to overemphasize short-term volatility while under-representing long-term trend information, causing transient rebounds to be erroneously identified as signs of price recovery. Upon reversal of the stock price, the RPE of most models becomes more significant. This is also consistent with the overfitting of short-term fluctuations as explained above.

In contrast, the XSI-BiLSTM model, which integrates investor sentiment, exhibits superior performance. This enhancement may be because people's judgment of future stock prices is reflected in textual sentiment, and accurate bias correction can be achieved by using textual sentiment, which allows the XSI-BiLSTM to achieve a

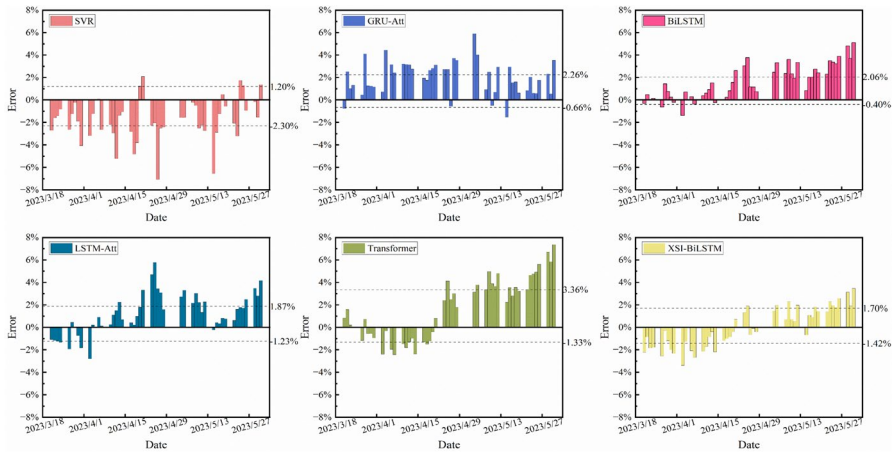


Fig. 9 RPE of each model for SZ399006

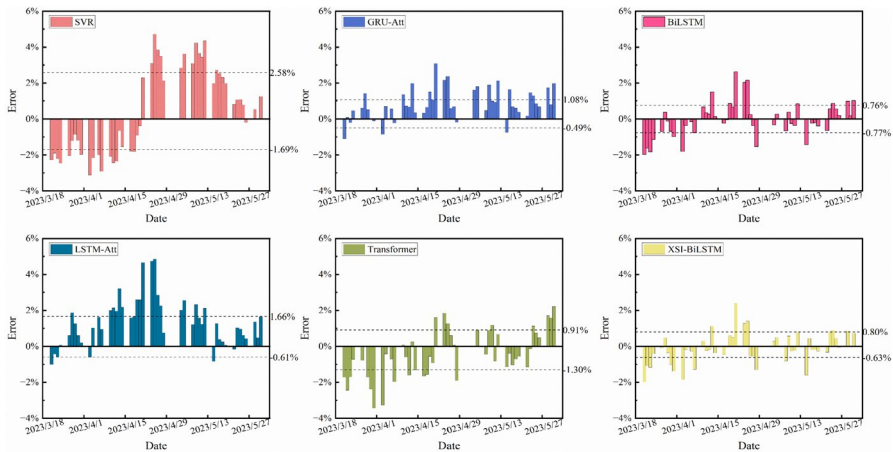


Fig. 10 RPE of each model for SZ399001

more balanced AOSE. In short, the stock price prediction task is highly complex, and predicting the price based on the time-series characteristics of the stock's technical indicators alone may not be accurate enough.

5.4 Ablation Experiment

In the prediction module of XSI-BiLSTM, we incorporate an investor sentiment index and introduce a highway mechanism to enhance the information-capturing ability of the BiLSTM model. Therefore, it is necessary to verify its effectiveness through ablation experiments. Tables 8 and 9 show the performance comparison results between BiLSTM, BiLSTM using sentiment index (SI), BiLSTM using the highway mecha-

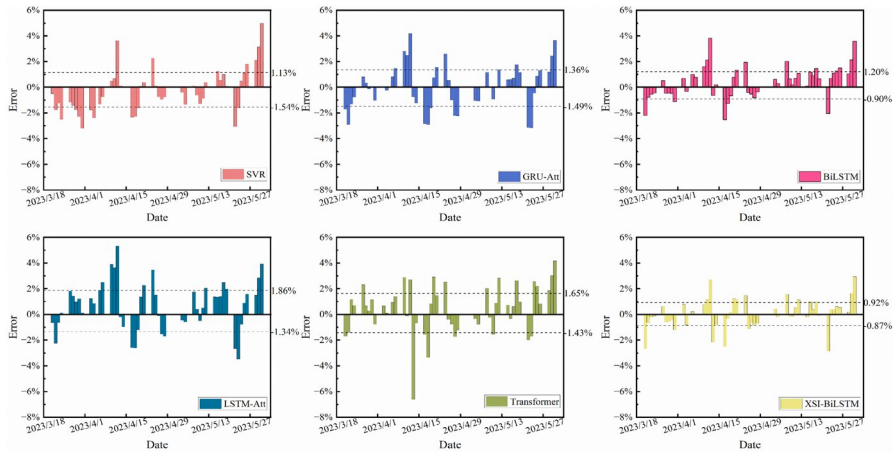


Fig. 11 RPE of each model for SZ600519

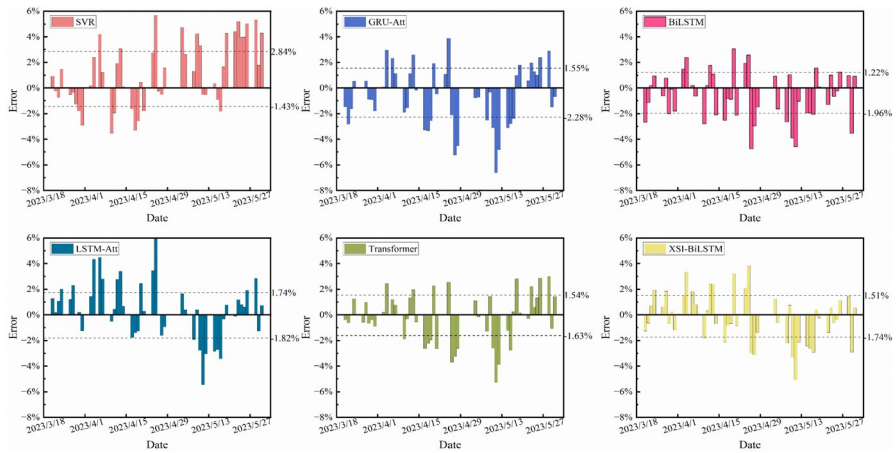


Fig. 12 RPE of each model for SZ300750

nism, and both (i.e., the modules we use). The results are presented in multiplicative form using BiLSTM performance as a benchmark.

Table 8 evaluates accuracy through RMSE and MAPE, quantifying the magnitude of prediction errors, while paired t-tests are conducted between each model variant and the benchmark to assess the statistical significance of improvements. Meanwhile, Table 9 presents R2, which measures the model's explanatory power regarding price variance. The results indicate that either mechanism individually enhances performance. We observed that under shorter input windows, the improvements were more effective and often more significant. Hence, Fig. 13 further illustrates the relative improvement of each variant across different input windows and metrics. The highway mechanism yields greater benefits under shorter input windows, confirming its efficiency in preserving information over short-term dependencies, while the senti-

Table 8 Performance comparison results of the ablation experiment (RMSE and MAPE)

Input windows		7			15			30		
Stock	Model	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	
	BiLSTM	1x	1x	1x	1x	1x	1x	1x	1x	
SZ399006	BiLSTM-SI	1.18x(1.89)	1.14x(1.36)	1.23x(1.74)	1.20x(1.76)	1.22x(2.29*)		1.14x(0.95)		
	BiLSTM-highway	1.29x(2.21*)	1.23x(2.51*)	1.04x(1.00)	1.01x(-0.04)	1.25x(4.47***)	1.21x(2.99**)			
	Our	1.35x(3.34**)	1.37x(1.57)	1.19(2.64*)	1.21x(0.86)	1.29x(2.22*)	1.20x(1.14)			
SZ399001	BiLSTM-SI	1.16x(1.68)	1.14x(1.21)	1.20x(1.94)	1.22x(1.71)	1.16x(2.50*)	1.15x(2.14*)			
	BiLSTM-highway	1.17x(1.52)	1.19x(1.07)	1.23x(2.30*)	1.16x(1.49)	1.11x(2.32*)	1.08x(1.65)			
	Our	1.24x(3.05**)	1.23x(2.60*)	1.24x(1.94)	1.19x(1.38)	1.13x(2.03*)	1.11x(1.60)			
SH600519	BiLSTM-SI	1.07x(1.14)	1.14x(1.63)	1.15x(1.52)	1.08x(0.72)	1.07x(0.94)	1.13x(1.48)			
	BiLSTM-highway	1.09x(1.38)	1.12x(1.66)	1.13x(1.79)	1.07x(0.72)	1.02x(0.24)	1.08x(0.72)			
	Our	1.13x(2.21*)	1.21x(2.61*)	1.15x(1.63)	1.10x(0.86)	1.14x(1.63)	1.23x(2.43*)			
SZ300750	BiLSTM-SI	1.12x(1.25)	1.14x(1.63)	1.10x(1.17)	1.11x(1.34)	0.98x(-0.47)	0.99x(-0.13)			
	BiLSTM-highway	1.12x(1.86)	1.09x(1.49)	1.05x(0.79)	1.03x(0.32)	1.02x(0.65)	1.02x(0.43)			
	Our	1.16x(2.04*)	1.16x(2.20*)	1.11x(0.98)	1.10x(1.06)	1.01x(0.18)	1.02x(0.40)			

Table 9 Performance comparison results of the ablation experiment (R2)

Stock	SZ399006				SZ399001				SH600519				SZ300750			
Model/Input windows	7	15	30	7	15	30	7	15	30	7	15	30	7	15	30	7
BiLSTM	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x	1x
BiLSTM-SI	1.16x	1.15x	1.26x	1.03x	1.04x	1.03x	1.01x	1.06x	1.00x	1.18x	1.17x	1.02x	1.14x	1.06x	1.03x	1.02x
BiLSTM-highway	1.25x	1.04x	1.31x	1.04x	1.05x	1.02x	1.07x	1.10x	1.02x	1.22x	1.18x	1.05x	1.07x	1.07x	1.06x	1.06x
Our	1.26x	1.12x	1.32x	1.04x	1.04x	1.02x	1.05x	1.07x	1.05x	1.22x	1.18x	1.06x	1.22x	1.18x	1.06x	1.06x

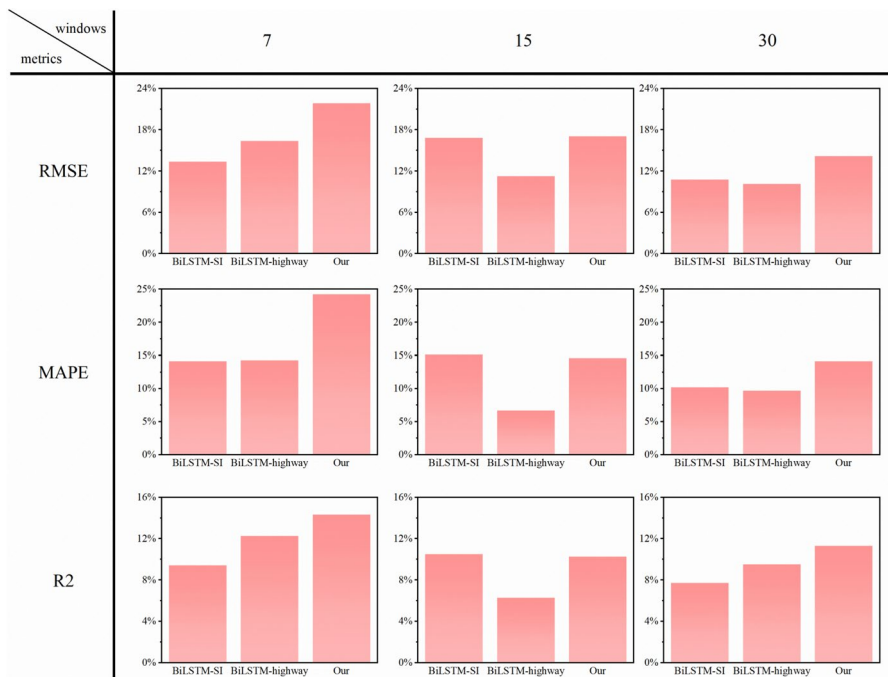


Fig. 13 Performance improvement effect of each model

ment index shows stronger advantages at medium and longer horizons by capturing persistent market sentiment. These findings imply that the two components operate in distinct yet complementary temporal domains. Overall, their joint integration produces the most balanced gains across accuracy and explanatory strength, reinforcing the architectural design rationale of XSI-BiLSTM.

In addition, Fig. 14 shows a box plot of the absolute error of prediction for each model. The errors show a skewed distribution, i.e., the difference between the model's predicted price and the actual price is negligible in most cases. The introduction of investor sentiment effectively reduces the median prediction error, and introducing the highway mechanism reduces the errors of extreme values. After combining the two approaches, the forecast error decreases overall, as reflected by the fact that its average error (i.e., the black square in the figure) is the lowest among all four stocks.

6 Conclusion

This study proposes XSI-BiLSTM, a hybrid framework for stock price prediction. Extensive experiments on two major Chinese indices (SZ399006, SZ399001) and two representative stocks (Kweichow Moutai and CATL) confirm its superiority. In sentiment analysis, the fine-tuned XLNET achieves an F1-score of 81.97; In predictive tasks, XSI-BiLSTM outperformed baseline models by at least 8.15%, 6.21%, and 5.80% in RMSE, MAPE, and R2, respectively.

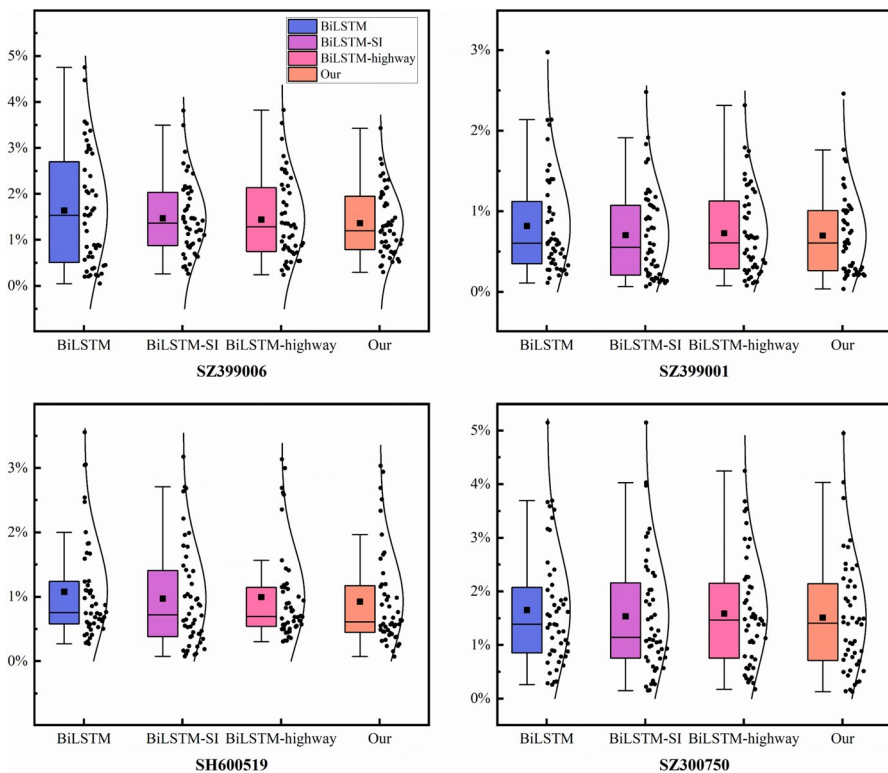


Fig. 14 Box-distribution plots of each model's absolute error

Our study focuses on the relatively understudied sentiment of retail investors rather than professional news sentiment. Additionally, we highlight the amplifying effect of popularity on sentiment, and the Granger causality tests validate that the popularity-weighted investor sentiment indices exhibit bidirectional relationships with stock price changes, confirming the enhanced effectiveness of sentiment-based forecasting. In XSI-BiLSTM, we are the first to apply XLNET and highway-enhanced BiLSTM to stock market forecasting, demonstrating their effectiveness.

Despite the promising results, several limitations of the XSI-BiLSTM framework should be acknowledged. First, as with many deep learning models, there is a risk of overfitting. Although we collected approximately one year of forum posts and conducted extensive experiments, overfitting remains a possibility. Second, the framework's performance is highly dependent on the quality and representativeness of online forum data. Biases like spam messages may distort the true investor sentiment signal. Although we have carried out screening and preprocessing on the collected forum data, this risk still exists. Finally, the framework has been evaluated only on Chinese indices and stocks, and its generalizability to other markets and asset classes remains to be systematically validated. Therefore, future research can address several key areas to expand its applicability and robustness:

- (1) **Cross-Market Generalization.** The current study primarily examines the Chinese stock markets, given the constraints imposed by data availability. Subsequent research aims to extend this analytical framework to other financial markets, particularly developed economies such as those in the US and EU, to assess the model's broader applicability across diverse financial systems.
- (2) **Multimodal Data Integration.** In addition to textual data, future work will extend this framework by incorporating multimodal data, such as emojis, shared images, and their cross-modal interactions with textual content.
- (3) **Real-Time Deployment.** Current research, including this study, primarily relies on daily or panel data. A critical direction is to enable real-time deployment in live trading environments. To achieve this goal, we will explore the application of reinforcement learning (Huang et al., 2024; Zou et al., 2024) in the prediction module of framework.
- (4) **Interpretability.** Given the complexity of deep learning architectures, enhancing the interpretability of the XSI-BiLSTM framework is crucial. Future research will aim to quantify the influence of investor sentiment and other input factors on real-time prediction outcomes. Interpretability techniques such as attention weight visualization or SHAP value analysis will be taken into consideration.
- (5) **Commentary Sentiment Integration.** This study focuses on the sentiment analysis of the original post, but each post also generates comments, which provide additional insights. Future work could integrate commentary sentiment with the original post. By analyzing their similarities and differences, enabling a deeper understanding of behavioral finance theories related to collective investor behavior transmission and information cascades (Bikhchandani et al., 2024).

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10614-025-11268-6>.

Author Contributions Huiyu Li: Conceptualization, Methodology, Data curation, Software, Writing - original draft. Junhua Hu: Funding acquisition, Project administration, Writing – review & editing.

Data Availability Data will be available on request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Aljifri, R. (2023). Investor psychology in the stock market: An empirical study of the impact of overconfidence on firm valuation. *Borsa Istanbul Review*, 23(1), 93–112. <https://doi.org/10.1016/j.bir.2022.09.010>
- Al-Khasawneh, M. A., Raza, A., Khan, S. U. R., & Khan, Z. (2025). Stock market trend prediction using deep learning approach. *Computational Economics*, 66(1), 453–484. <https://doi.org/10.1007/s10614-024-10714-1>

- Althelaya, K. A., El-Alfy, E.-S. M., & Mohammed, S. (2018). Evaluation of bidirectional LSTM for short- and long-term stock market prediction. In *2018 9th international conference on information and communication systems (ICICS)* (pp. 151–156). <https://doi.org/10.1109/IACS.2018.8355458>
- An, W., Wang, L., & Zeng, Y.-R. (2024). Social media-based multi-modal ensemble framework for forecasting soybean futures price. *Computers and Electronics in Agriculture*, 226, 109439. <https://doi.org/10.1016/j.compag.2024.109439>
- Anderson, L. R., & Holt, C. A. (1997). Information cascades and social learning. *The American Economic Review*, 87(5), 847–862. <https://www.jstor.org/stable/2951328>
- Antweiler, W., & Frank, M. Z. (2004). Is all that talk just noise? The information content of internet stock message boards. *The Journal of Finance*, 59(3), 1259–1294. <https://doi.org/10.1111/j.1540-6261.2004.00662.x>
- Ariyo, A. A., Adewumi, A. O., & Ayo, C. K. (2014). Stock price prediction using the ARIMA model. In *2014 UKSim-AMSS 16th international conference on computer modelling and simulation* (pp. 106–112). <https://doi.org/10.1109/UKSim.2014.67>
- Asghar, M. Z., Rahman, F., Kundi, F. M., & Ahmad, S. (2019). Development of stock market trend prediction system using multiple regression. *Computational and Mathematical Organization Theory*, 25(3), 271–301. <https://doi.org/10.1007/s10588-019-09292-7>
- Barber, B. M., Huang, X., Odean, T., & Schwarz, C. (2022). Attention-induced trading and returns: Evidence from Robinhood users. *Journal of Finance*, 77(6), 3141–3190. <https://doi.org/10.1111/jofi.13183>
- Beniwal, M., Singh, A., & Kumar, N. (2023). Forecasting long-term stock prices of global indices: A forward-validating genetic algorithm optimization approach for support vector regression. *Applied Soft Computing*, 145, Article 110566. <https://doi.org/10.1016/j.asoc.2023.110566>
- Bikhchandani, S., Hirshleifer, D., Tamuz, O., & Welch, I. (2024). Information cascades and social learning. *Journal of Economic Literature*, 62(3), 1040–1093. <https://doi.org/10.1257/jel.20241472>
- Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5), 992–1026. <https://doi.org/10.1086/261849>
- Costola, M., Hinz, O., Nofer, M., & Pelizzon, L. (2023). Machine learning sentiment analysis, COVID-19 news and stock market reactions. *Research in International Business and Finance*, 64, Article 101881. <https://doi.org/10.1016/j.ribaf.2023.101881>
- Dai, Y., Wang, G., Dai, J., & Geman, O. (2020). A multimodal deep architecture for traditional Chinese medicine diagnosis. *Concurrency and Computation: Practice & Experience*, 32(19), Article e5781. <https://doi.org/10.1002/cpe.5781>
- Daniel, K., & Titman, S. (1999). Market efficiency in an irrational world. *Financial Analysts Journal*, 55(6), 28–40. <https://doi.org/10.2469/faj.v55.n6.2312>
- Das, N., Sadhukhan, B., Ghosh, R., & Chakrabarti, S. (2024). Developing hybrid deep learning models for stock price prediction using enhanced twitter sentiment score and technical indicators. *Computational Economics*, 64(6), 3407–3446. <https://doi.org/10.1007/s10614-024-10566-9>
- Dong, J. (2023). Natural language processing pretraining language model for computer intelligent recognition technology. *ACM Transactions on Asian and Low-Resource Language Information Processing*, 23(8), 1–12. <https://doi.org/10.1145/3605210>
- Doroshenko, L., & Tu, F. (2023). Like, share, comment, and repeat: Far-right messages, emotions, and amplification in social media. *Journal of Information Technology & Politics*, 20(3), 286–302. <https://doi.org/10.1080/19331681.2022.2097358>
- Eachempati, P., Srivastava, P. R., Kumar, A., de Muñoz Prat, J., & Delen, D. (2022). Can customer sentiment impact firm value? An integrated text mining approach. *Technological Forecasting and Social Change*, 174, Article 121265. <https://doi.org/10.1016/j.techfore.2021.121265>
- Edmans, A., Fernandez-Perez, A., Garel, A., & Indriawan, I. (2022). Music sentiment and stock returns around the world. *Journal of Financial Economics*, 145(2, Part A), 234–254. <https://doi.org/10.1016/j.jfineco.2021.08.014>
- Fama, E. F. (1991). Efficient Capital Markets: II. *The Journal of Finance*, 46(5), 1575–1617. <https://doi.org/10.1111/j.1540-6261.1991.tb04636.x>
- Gite, S., Khatavkar, H., Kotecha, K., Srivastava, S., Maheshwari, P., & Pandey, N. (2021). Explainable stock prices prediction from financial news articles using sentiment analysis. *PeerJ Computer Science*, 7, Article e340. <https://doi.org/10.7717/peerj-cs.340>
- Gong, J., & Sun, S. (2009). A new approach of stock price prediction based on logistic regression model. *International Conference on New Trends in Information and Service Science*, 2009, 1366–1371. <https://doi.org/10.1109/NISS.2009.267>

- Gong, J., Wang, G.-J., Xie, C., & Uddin, G. S. (2024). How do market volatility and risk aversion sentiment inter-influence over time? Evidence from Chinese SSE 50 ETF options. *International Review of Financial Analysis*, 95, Article 103440. <https://doi.org/10.1016/j.irfa.2024.103440>
- Gong, X., Zhang, W., Wang, J., & Wang, C. (2022). Investor sentiment and stock volatility: New evidence. *International Review of Financial Analysis*, 80, Article 102028. <https://doi.org/10.1016/j.irfa.2022.102028>
- Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438. <https://doi.org/10.2307/1912791>
- Han, X., Zhang, Z., Ding, N., Gu, Y., Liu, X., Huo, Y., Qiu, J., Yao, Y., Zhang, A., Zhang, L., Han, W., Huang, M., Jin, Q., Lan, Y., Liu, Y., Liu, Z., Lu, Z., Qiu, X., Song, R., et al. (2021). Pre-trained models: Past, present and future. *AI Open*, 2, 225–250. <https://doi.org/10.1016/j.aiopen.2021.08.002>
- Huang, C., Cao, Y., Lu, M., Shan, Y., & Zhang, Y. (2023). Messages in online stock forums and stock price synchronicity: Evidence from China. *Accounting & Finance*, 63(3), 3011–3041. <https://doi.org/10.1111/acfi.13005>
- Huang, J.-Y., Tung, C.-L., & Lin, W.-Z. (2023). Using social network sentiment analysis and genetic algorithm to improve the stock prediction accuracy of the deep learning-based approach. *International Journal of Computational Intelligence Systems*, 16(1), Article 93. <https://doi.org/10.1007/s44196-023-00276-9>
- Huang, Y., Zhou, C., Cui, K., & Lu, X. (2024). A multi-agent reinforcement learning framework for optimizing financial trading strategies based on TimesNet. *Expert Systems with Applications*, 237, Article 121502. <https://doi.org/10.1016/j.eswa.2023.121502>
- Jin, Z., Yang, Y., & Liu, Y. (2020). Stock closing price prediction based on sentiment analysis and LSTM. *Neural Computing and Applications*, 32(13), 9713–9729. <https://doi.org/10.1007/s00521-019-04504-2>
- Johnman, M., Vanstone, B. J., & Gepp, A. (2018). Predicting FTSE 100 returns and volatility using sentiment analysis. *Accounting & Finance*, 58(S1), 253–274. <https://doi.org/10.1111/acfi.12373>
- Kanwal, A., Lau, M. F., Ng, S. P. H., Sim, K. Y., & Chandrasekaran, S. (2022). BiCuDNNLSTM-1dCNN — A hybrid deep learning-based predictive model for stock price prediction. *Expert Systems with Applications*, 202, Article 117123. <https://doi.org/10.1016/j.eswa.2022.117123>
- Khan, S., & Alghulaiaikh, H. (2020). ARIMA model for accurate time series stocks forecasting. *International Journal of Advanced Computer Science and Applications*, 11(7), 524–528. <https://doi.org/10.14569/IJACSA.2020.0110765>
- Kim, C., & Yang, S.-U. (2017). Like, comment, and share on Facebook: How each behavior differs from the other. *Public Relations Review*, 43(2), 441–449. <https://doi.org/10.1016/j.pubrev.2017.02.006>
- Lee, W. Y., Jiang, C. X., & Indro, D. C. (2002). Stock market volatility, excess returns, and the role of investor sentiment. *Journal of Banking & Finance*, 26(12), 2277–2299. [https://doi.org/10.1016/S0378-4266\(01\)00202-3](https://doi.org/10.1016/S0378-4266(01)00202-3)
- Li, X., Xie, H., Chen, L., Wang, J., & Deng, X. (2014). News impact on stock price return via sentiment analysis. *Knowledge-Based Systems*, 69, 14–23. <https://doi.org/10.1016/j.knosys.2014.04.022>
- Lin, P., Ma, S., & Fildes, R. (2024). The extra value of online investor sentiment measures on forecasting stock return volatility: A large-scale longitudinal evaluation based on Chinese stock market. *Expert Systems with Applications*, 238, Article 121927. <https://doi.org/10.1016/j.eswa.2023.121927>
- Lin, Y.-L., Lai, C.-J., & Pai, P.-F. (2022). Using deep learning techniques in forecasting stock markets by hybrid data with multilingual sentiment analysis. *Electronics*, 11(21), 3513. <https://doi.org/10.3390/electronics11213513>
- Liu, Q., Lee, W.-S., Huang, M., & Wu, Q. (2023). Synergy between stock prices and investor sentiment in social media. *Borsa Istanbul Review*, 23(1), 76–92. <https://doi.org/10.1016/j.bir.2022.09.006>
- Liu, W.-J., Ge, Y.-B., & Gu, Y.-C. (2024). News-driven stock market index prediction based on trellis network and sentiment attention mechanism. *Expert Systems with Applications*, 250, Article 123966. <https://doi.org/10.1016/j.eswa.2024.123966>
- Luo, S., Shyu, M.-L., Silalahi, A., Dayarana, K., & Shih, I.-T. (2022). A business model predicting the impact of live-stream marketing on firms' stock price—An event study. *Heliyon*, 8(12). <https://doi.org/10.1016/j.heliyon.2022.e12522>
- Mishev, K., Gjorgjevikj, A., Vodenska, I., Chitkushev, L. T., & Trajanov, D. (2020). Evaluation of sentiment analysis in finance: From lexicons to transformers. *IEEE Access*, 8, 131662–131682. <https://doi.org/10.1109/ACCESS.2020.3009626>

- Mu, G., Gao, N., Wang, Y., & Dai, L. (2023). A stock price prediction model based on investor sentiment and optimized deep learning. *IEEE Access*, 11, 51353–51367. <https://doi.org/10.1109/ACCESS.2023.3278790>
- Nikmah, C., & Hung, R.-J. (2024). The impact of ASEAN economic community, firm characteristics and macroeconomics on firm performance and firm value: An investigation of Shariah-compliant firms in Indonesia. *Heliyon*. <https://doi.org/10.1016/j.heliyon.2024.e32740>
- Niu, H., & Xu, K. (2020). A hybrid model combining variational mode decomposition and an attention-GRU network for stock price index forecasting. *Mathematical Biosciences and Engineering*, 17(6), 7151–7166. <https://doi.org/10.3934/mbe.2020367>
- Obaid, K., & Pukthuanthong, K. (2022). A picture is worth a thousand words: Measuring investor sentiment by combining machine learning and photos from news. *Journal of Financial Economics*, 144(1), 273–297. <https://doi.org/10.1016/j.jfineco.2021.06.002>
- Park, H. J., Kim, Y., & Kim, H. Y. (2022). Stock market forecasting using a multi-task approach integrating long short-term memory and the random forest framework. *Applied Soft Computing*, 114, Article 108106. <https://doi.org/10.1016/j.asoc.2021.108106>
- Pedersen, L. H. (2022). Game on: Social networks and markets. *Journal of Financial Economics*, 146(3), 1097–1119. <https://doi.org/10.1016/j.jfineco.2022.05.002>
- Reis, P. M. N., & Pinho, C. (2021). A reappraisal of the causal relationship between sentiment proxies and stock returns. *Journal of Behavioral Finance*, 22(4), 420–442. <https://doi.org/10.1080/15427560.2020.1792910>
- Rezaei, H., Faaljou, H., & Mansourfar, G. (2021). Stock price prediction using deep learning and frequency decomposition. *Expert Systems with Applications*, 169, Article 114332. <https://doi.org/10.1016/j.eswa.2020.114332>
- Sapankevych, N. I., & Sankar, R. (2009). Time series prediction using support vector machines: A survey. *IEEE Computational Intelligence Magazine*, 4(2), 24–38. <https://doi.org/10.1109/MCI.2009.932254>
- Sevinç, D., & Coşkun, M. (2025). The impact of social media and internet forums posts on the stock market dynamics: A sentiment analysis using a lexical approach in Borsa İstanbul. *Applied Economics*, 58, 1–13. <https://doi.org/10.1080/00036846.2025.2486791>
- Shanthini, P. M., Parthasarathy, S., Venkatesan, P., & Nandhini, S. (2023). HRSR-SVM: Hybrid reptile search Remora-based support vector machine for forecasting stock price movement. *International Journal of Information Technology*, 15(6), 3127–3134. <https://doi.org/10.1007/s41870-023-01331-6>
- Sharaf, M., Hemdan, E.E.-D., El-Sayed, A., & El-Bahnasawy, N. A. (2021). StockPred: A framework for stock price prediction. *Multimedia Tools and Applications*, 80(12), 17923–17954. <https://doi.org/10.1007/s11042-021-10579-8>
- Sweidan, A. H., El-Bendary, N., & Al-Feel, H. (2021). Sentence-level aspect-based sentiment analysis for classifying adverse drug reactions (ADRs) using hybrid ontology-XLNet transfer learning. *IEEE Access*, 9, 90828–90846. <https://doi.org/10.1109/ACCESS.2021.3091394>
- Valle-Cruz, D., Fernandez-Cortez, V., López-Chau, A., & Sandoval-Almazán, R. (2022). Does twitter affect stock market decisions? Financial sentiment analysis during pandemics: A comparative study of the H1N1 and the COVID-19 periods. *Cognitive Computation*, 14(1), 372–387. <https://doi.org/10.1007/s12559-021-09819-8>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 31, 6000–6010. <https://doi.org/10.5555/3295222.3295349>
- Wang, M., Wang, S., Guo, J., & Jia, W. (2024). Improving stock trend prediction with pretrain multi-granularity denoising contrastive learning. *Knowledge and Information Systems*, 66(4), 2439–2466. <https://doi.org/10.1007/s10115-023-02006-1>
- Wang, Y., & Guo, Y. (2020). Forecasting method of stock market volatility in time series data based on mixed model of ARIMA and XGBoost. *China Communications*, 17(3), 205–221. <https://doi.org/10.23919/JCC.2020.03.017>
- Wu, S., Liu, Y., Zou, Z., & Weng, T.-H. (2022). S_I_LSTM: Stock price prediction based on multiple data sources and sentiment analysis. *Connection Science*, 34(1), 44–62. <https://doi.org/10.1080/09540091.2021.1940101>
- Xu, Y., Liang, C., Li, Y., & Huynh, T. L. D. (2022). News sentiment and stock return: Evidence from managers' news coverages. *Finance Research Letters*, 48, Article 102959. <https://doi.org/10.1016/j.frl.2022.102959>

- Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R. R., & Le, Q. V. (2019). XLNet: Generalized Autoregressive Pretraining for Language Understanding. *Advances in Neural Information Processing Systems*, 32, 5753–5763. <https://doi.org/10.5555/3454287.3454804>
- Yun, K. K., Yoon, S. W., & Won, D. (2021). Prediction of stock price direction using a hybrid GA-XGBoost algorithm with a three-stage feature engineering process. *Expert Systems with Applications*, 186, Article 115716. <https://doi.org/10.1016/j.eswa.2021.115716>
- Zhang, X., Liang, X., Zhiyuli, A., Zhang, S., Xu, R., & Wu, B. (2019). At-lstm: An attention-based lstm model for financial time series prediction. *IOP Conference Series: Materials Science and Engineering*, 569(5), Article 052037. <https://doi.org/10.1088/1757-899X/569/5/052037>
- Zilly, J. G., Srivastava, R. K., Koutnik, J., & Schmidhuber, J. (2017). Recurrent highway networks. In *Proceedings of the 34th international conference on machine learning* (pp. 4189–4198). <https://doi.org/10.5555/3305890.3306114>
- Zou, J., Lou, J., Wang, B., & Liu, S. (2024). A novel deep reinforcement learning based automated stock trading system using cascaded LSTM networks. *Expert Systems with Applications*, 242, Article 122801. <https://doi.org/10.1016/j.eswa.2023.122801>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.